Set: A

CSE 423 Quiz -1

Mark: 20, Time: 20 minutes

Name: ID: Section:

1. What is aliasing? Explain briefly. 3

Slide

2. Find the first 6 pixels of the line (-5, 0) to (4, -7) using Midpoint line algorithm. 8

$dx = 4+5 = 9 > 0$

$dy = -7-0 = -7 < 0$

$|dx| > |dy|$ ∴ Zone 7

Zone 0 to Zone 7

$X = x, \quad Y = -y$

Zone 7 to Zone 0:

$X = x, \quad Y = -y$

$(-5, 0) \rightarrow (-5, 0)$

$(4, -7) \rightarrow (4, 7)$

$dx' = 4+5 = 9$

$dy' = 7-0 = 7$

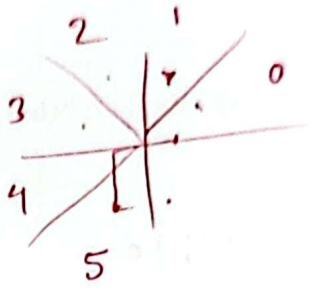
$D = 2\times7 - 9 = 5$

$\Delta NE = 2(7-9) = -4$

$\Delta E = 2\times7 = 14$

| x' | y' | d | NE/E | Update D | x | y |
|---|---|---|---|---|---|---|
| -5 | 0 | 5>0 | NE | 1 | -5 | 0 |
| -4 | 1 | 1>0 | NE | -3 | -4 | -1 |
| -3 | 2 | -3<0 | E | 11 | -3 | -2 |
| -2 | 2 | 11>0 | NE | 7 | -2 | -2 |
| -1 | 3 | 7>0 | NE | 3 | -1 | -3 |
| 0 | 4 | 3>0 | NE | -1 | 0 | -4 |
| 1 | 5 | -1<0 | E | 13 | 1 | -5 |

3. Write the rule to convert Zone 5 to Zone 1. 3




~~X = -y, Y = -x~~

X = -x, Y = -y

4. Using DDA algorithm find the first 4 pixels of the line segment from (0, 3) to (4, 9) 4

$m = \frac{9-3}{4-0} = \frac{6}{4} = \frac{3}{2} = 1.5$ $\quad \frac{1}{m} = 0.67$

| y | x | x |
|---|---|---|
| 3 | 0 | 0 |
| 4 | 0.67 | 1 |
| 5 | 1.34 | 1 |
| 6 | 2.01 | 2 |
| 7 | 2.68 | 3 |

5. True/False 2

a. Virtual Reality (VR) projects information into the real world. F

b. Raster displays do not have aliasing problems. F

CamScanner

CSE 423 Quiz -1

Set: B

Mark: 20, Time: 20 minutes

Name: ID: Section:

1. Briefly explain what a frame buffer is. 3

Slide

2. Using DDA algorithm find the first 4 pixels of the line segment from (9, 3) to (3, 0) 4

$$m = \frac{0-3}{3-9} = \frac{1}{2}$$

$-1 < m < 1$

$\therefore x_0 = 9 > x_1 = 3$

$\therefore x_{k+1} = x_k - 1$

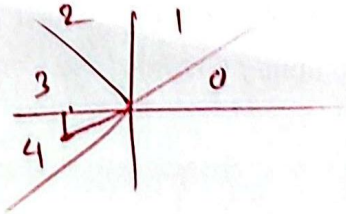
$y_{k+1} = y_k - m$

| x | y | y(round) |
|---|---|---|
| 9 | 3 | 3 |
| 8 | 2.5 | 3 |
| 7 | 2 | 2 |
| 6 | 1.5 | 2 |
| 5 | 1 | 1 |

3. Write the conditions to identify Zone 4. 3

$dx < 0, \; dy < 0$

$|dx| > |dy|$




CamScanner

4. Find the first 6 pixels of the line (3, 4) to (-10, -7) using Midpoint line algorithm. 8

$dx = -10-3 = -13 < 0$

$dy = -7-4 = -11 < 0$

$|dx| > |dy|$

∴ Zone 4

Zone 4 to Zone 0:

$X = -x,\ Y = -y$

$(3, 4) \rightarrow (-3, -4)$

$(-10, -7) \rightarrow (10, 7)$

$dx' = 10+3 = 13$

$dy' = 7+4 = 11$

$D = 2 \times 11 - 13 = 9$

$\Delta NE = 2(11-13) = -4$

$\Delta E = 2 \times 11 = 22$

Zone 0 to Zone 4

$X = -x,\ Y = -y$

| X' | Y' | D | | $Update_D$ | X | Y |
|---|---|---|---|---|---|---|
| -3 | -4 | 9>0 | NE | 5 | 3 | 4 |
| -2 | -3 | 5>0 | NE | 1 | 2 | 3 |
| -1 | -2 | 1>0 | NE | -3 | 1 | 2 |
| 0 | -1 | -3<0 | E | 19 | 0 | 1 |
| 1 | -1 | 19>0 | NE | 15 | -1 | 1 |
| 2 | 0 | 15>0 | NE | 11 | -2 | 0 |
| 3 | 1 | 11>0 | NE | 7 | -3 | -1 |
| | | | | | | |

5. True/False 2

a. Raster displays do not have aliasing problems. F

b. Computer Graphics and Computer Vision are the same thing. F

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CSE 423 Quiz -1

Set: C

Mark: 20, Time: 20 minutes

Name: ID: Section:

1. Increasing screen resolution reduces aliasing. But why is this method usually avoided? 3

Slide

2. Find the first 6 pixels of the line (-5, 0) to (4, -8) using Midpoint line algorithm. 8

$dx = 4+5 = 9 > 0$

$dy = -8-0 = -8 < 0$

$|dx| > |dy|$ ∴ Zone 7

Zone 7 to Zone 0: $X = x,\ Y = -y$

$(-5, 0) \rightarrow (-5, 0)$

$(4, -8) \rightarrow (4, 8)$

$dx = 4+5 = 9$

$dy = 8$

$D = 2\times 8 - 9 = 7$

$\Delta NE = 2\times(8-9) = -2$

$\Delta E = 2\times 8 = 16$

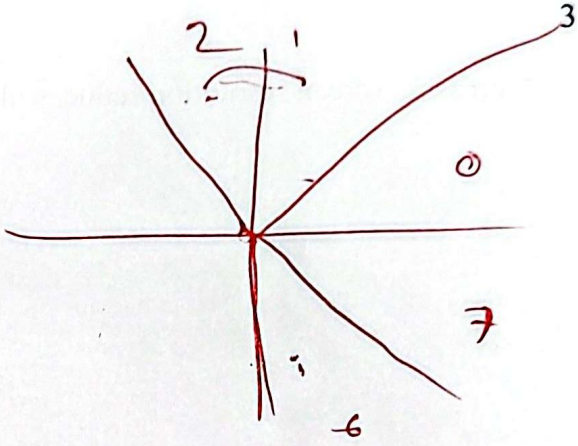
| x | y | D | Update d | x' | y' |
|---|---|---|---|---|---|
| -5 | 0 | 7>0 | 5 | -5 | 0 |
| -4 | 1 | 5>0 | 3 | -4 | -1 |
| -3 | 2 | 3>0 | 1 | -3 | -2 |
| -2 | 3 | 1>0 | -2 | -2 | -3 |
| -1 | 4 | -1<0 | 15 | -1 | -4 |
| 0 | 4 | 15>0 | 13 | 0 | -4 |
| 1 | 5 | 13>0 | 11 | 1 | -5 |

Zone 0 to 7

$x = x$

$y = -Y$

3. Write the rule to convert Zone 6 to Zone 2. 3




$6 \rightarrow 1$: $x' = X$, $y' = -Y$

$1 \rightarrow 2$: $x = -x'$, $y = y'$

$6 \rightarrow 2$: $x = -X$, $y = -Y$

4. Using DDA algorithm find the first 4 pixels of the line segment from (9, 3) to (3, 0) 4

| x | y | y (round) |
|---|---|---|
| 9 | 3 | 3 |
| 8 | 2.5 | 3 |
| 7 | 2 | 2 |
| 6 | 1.5 | 2 |
| 5 | 1 | 1 |

$m = \frac{0-3}{3-9} = \frac{1}{2} = 0.5$

$x_0 = 9 > x_1 = 3$

$\therefore x_{k+1} = x_k - 1$

$y_{k+1} = y_k - m$

5. True/False 2

a. Computer Graphics and Computer Vision are the same thing. F

b. Aliasing cannot be completely eliminated. T

Set: D

CSE 423 Quiz -1

Mark: 20, Time: 20 minutes

Name: ID: Section:

1. Mention some limitations of DDA algorithm. 3

slide

2. Using DDA algorithm find the first 4 pixels of the line segment from (0, 3) to (4, 8) 4

$m = \frac{8-3}{4-0} = \frac{5}{4} = 1.25$

| y | x | x | y |
|---|---|---|---|
| 3 | 0 | 0 | 3 |
| 4 | .8 | 1 | 4 |
| 5 | 1.6 | 2 | 5 |
| 6 | 2.4 | 2 | 6 |
| 7 | 3.2 | 3 | 7 |

Round

~~$x_0 = 0$~~

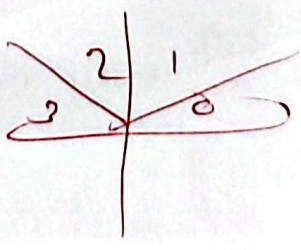
$y_0 = 3 < y_1 = 8$

$\therefore y_{k+1} = y_k + 1$

$x_{k+1} = x_k + \frac{1}{m}$

$\frac{1}{m} = 0.8$

3. Write the conditions to identify Zone 3. 3




$dx < 0 \quad dy > 0$

$|dx| > |dy|$

4. Find the first 6 pixels of the line (3, 4) to (-10, -7) using Midpoint line algorithm. 8

$dx = -10-3 = -13 < 0$

$dy = -7-4 = -11 < 0$

$|dx| > |dy|$ ∴ zone 4

Zone 4 to Zone 0: $x = -X$, $y = -Y$

$(3, 4) \rightarrow (-3, -4)$

$(-10, -7) \rightarrow (10, 7)$

$dx' = 10+3 = 13$

$dy' = 7+4 = 11$

$D = 2 \times 11 - 13 = 9$

$\Delta NE = 2 \times (11-13) = -4$

$\Delta E = 2 \times 11 = 22$

back to Zone 4

$x = -X$

$y = -Y$

| x | y | D | update | x' | y' |
|---|---|---|---|---|---|
| -3 | -4 | 9>0 | 5 | 3 | 4 |
| -2 | -3 | 5>0 | 1 | 2 | 3 |
| -1 | -2 | 1>0 | -3 | 1 | 2 |
| 0 | -1 | -3<0 | 19 | 0 | 1 |
| 1 | -1 | 19>0 | 15 | -1 | 1 |
| 2 | 0 | 15>0 | 11 | -2 | 0 |
| 3 | 1 | 11 | | -3 | -1 |

5. True/False 2

a. Virtual Reality (VR) projects information into the real world. F

b. Aliasing can be completely eliminated. F

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